

Steer on a Sphere: Geometric Control of Transformer Outputs

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Abstract

Layer normalization constrains transformer hidden states to a sphere whose radius is $\|\gamma_l\|$ per layer (not the textbook $\sqrt{d}$). Within this sphere, a single tangent step toward any token’s LM-head direction increases that token’s logit with 100% reliability, and reaches rank 1 with high but model-dependent probability (97–99% on Qwen2-0.5B and Qwen2-1.5B, 91% on GPT-2, 67% on SmolLM-135M, 28% on Pythia-160M; all under identical controlled conditions). Corrected controls show that the result is target specific and that rank acquisition is governed by the LM-head decision partition, not only by alignment with the target row. For every reachable case, the target that becomes rank 1 along the tangent arc remains rank 1 at the arc endpoint (no mid-arc loss). We propose a theory of edge-of-chaos dynamics: the per-model Lyapunov rate clusters near $\lambda \cdot L_{\text{comp}} \approx 0.5$ for the Gemma, GPT-2, and mid-Qwen families. The BOS token defines a growing reference axis orthogonal to vocabulary tokens across layers. Some tokens are self-reinforcing fixed points (cow tipping)—the degenerate limit of the dynamics at $\lambda \rightarrow 0$. On small models these loops are decoder-dependent; on Qwen2.5-7B the strict 0 pit survives greedy and standard nucleus sampling, but is broken by the chat template. We derive a threat model and evaluate mitigations (repetition detection, entropy monitoring, n-gram and periodicity detectors, pit-away steering, input sanitization, output collapse, and decoder changes), reporting loop-break rates and false-positive rates on ordinary generation. Natural activations are confirmed to lie near a spherical shell (norm std $\approx 7\%$ of mean), and controlled tangent steps are substantially shorter than the actual steps the model takes when the target token is appended. Because pits are derived from model weights, they can be computed for any architecture without retraining or data.

1 The Sphere

RMSNorm maps hidden states to a sphere whose expected radius is $\|\gamma_l\|$:

$$\|\text{RMSNorm}(h)\|^2 = \sum_{i=1}^d \gamma_{l,i}^2 \left(\frac{h_i}{\text{RMS}(h)} \right)^2$$

Per-instance norms float around this value (ratio 0.7–1.3), so the picture is a fuzzy shell. Measured γ norms across four model families:

Model	d	$\ \gamma_0\ $	$\ \gamma_{\text{mid}}\ $	$\ \gamma_{\text{end}}\ $
Mistral-7B	4096	16.6	111–158	195.5
DeepSeek-7B	4096	3.7	15–24	26.1
Qwen2-1.5B	1536	20.0	40–60	71.0
Qwen2.5-7B	3584	17.0	63.5	90.0

$\|\gamma_l\|$ is learned, varying across architectures even at the same d .

2 BOS Axis

The BOS token (position 0) defines a reference direction $\hat{b}_l = h_l^{(0)} / \|h_l^{(0)}\|$. Its residual norm grows from 2.8 to 102.2 across 28 layers, consistent with attention-sinking [3]. LM head rows W_t are approximately orthogonal to BOS at every layer ($\theta_{l,t} = \arccos|\hat{b}_l \cdot \hat{w}_t|$):

Layer	$\ \text{LN}(h)\ $	$\ \text{BOS}\ $	Token–BOS angle
0	15.9	2.8	88–90°
2	16.3	24.8	88–90°
4	33.5	21.8	89–90°
6	31.8	36.9	89–90°
8	36.5	44.1	88–90°
10	34.7	43.4	89–90°
12	37.5	50.3	89–90°
14	42.9	55.2	88–90°
16	44.4	53.0	89–90°
18	40.7	43.9	89–90°
20	45.2	48.2	89–90°
22	54.9	63.7	89–89°
24	55.5	68.7	87–90°
26	55.6	83.1	86–90°
27	63.1	102.2	87–90°

High-dimensional vectors are generically near-orthogonal ($\mathbb{E}[\|\cos\|] \approx 1/\sqrt{d}$), so the approximate 90° is expected; the consistent preservation of this angle while BOS grows 36× across layers is the observed regularity. Mistral-7B and DeepSeek-7B also give 87–90°.

3 Sphere Steering: Tangent Traversal

The LM head row W_t has a tangent component $g_t = W_t - (W_t \cdot \hat{h})\hat{h}$. Moving h along g_t increases the target logit ($W_t \cdot g_t = \|g_t\|^2 \geq 0$). Renormalizing keeps the traversal on the sphere:

$$h' = \frac{h + \alpha g_t}{\|h + \alpha g_t\|} \cdot \|\gamma_L\|$$

Algebraically, tangent step plus renormalization is identical to a same-target great-circle rotation, so the angular budget can be made explicit ($\alpha = 0.3$ corresponds to at most $\arctan(0.3) \approx 16.7^\circ$).

3.1 Target Specificity

A natural control is to steer toward a different target while measuring the original target’s rank. In a batched implementation the wrong target must be sampled from the other $K - 1$ targets; allowing self-draws artificially inflates the wrong-target success rate. With the self-exclusion fix, the wrong-target rank-1 rate is exactly 0% on Qwen2-0.5B (1,024 cases), Qwen2-1.5B (1,024 cases), and Gemma-3-1B (2,048 cases). Random tangent directions also produce 0% target rank-1. The tangent direction is therefore genuinely target specific.

3.2 Competitor Geometry at Fixed Target Score

Reaching rank 1 is not the same as increasing alignment with the target row. The target tangent raises the target logit, but rank 1 requires the target to beat every competing row simultaneously. To isolate the competitor-direction variable, we decompose the target-tangent endpoint as $v_0 = \gamma s + \rho r$ with $r \perp s$ and rotate only the residual component:

$$v(\epsilon) = \gamma s + \rho(\cos \epsilon r + \sin \epsilon q), \quad q \perp s, \quad q \perp r.$$

This preserves both $\|v\|$ and the target score $v \cdot s$ (and hence the raw target logit). On Qwen2-0.5B (64 targets, 4 contexts, 4 layers, 8° residual rotation):

Condition	Target rank-1
Target-tangent endpoint	95.8%
Random residual direction	96.2%
Residual toward strongest blocker	15.9%
Residual away from strongest blocker	100.0%

The useful variable is competitor direction, not merely “being off the target arc.” The target tangent remains a natural default direction for increasing alignment with the target row, but rank acquisition is governed by the full LM-head decision partition.

3.3 Shortest Route to the Rank-1 Region

For the “how close is the target arc to the shortest route?” question, we project the unit state u onto the actual rank-1 cone defined by the raw LM-head rows:

$$C_t = \{x : (W_t - W_j)^\top x \geq 0 \ \forall j \neq t\}.$$

The projection is solved as an active-set Euclidean projection with dual non-negative least squares, giving a full-vocabulary feasibility certificate. For every pair where the target-row arc reaches rank 1, the cone distance satisfies $\theta_{\text{cell}} \leq \theta_{\text{author}} + \text{tol}$. On 24 targets $\times$ 2 contexts:

Model	θ_{author}	θ_{cell}	ratio
Qwen2-0.5B	10.02°	8.45°	1.25×
Gemma-3-1B	14.20°	10.28°	1.37×

The target-row arc is a principled and often efficient default route, but competitor-aware routes to the actual rank-1 region can be materially shorter. This gives an observable distinction between target-alignment efficiency and rank-acquisition efficiency.

3.4 Natural Activation Manifold

Natural hidden states at a fixed layer lie near a spherical shell whose radius is the learned RMSNorm scale. Measured on 52 ordinary prompts at layer 8 of Qwen2-0.5B, the post-layer-norm hidden-state norm has mean 13.81 and standard deviation 0.96 ($\approx 7\%$ of the mean), confirming the spherical approximation. Principal-component analysis shows the manifold is high-dimensional: the top 16 PCs explain only 56% of the variance.

When the model appends a target token, the resulting hidden-state step is much larger than a controlled tangent steering step of the same angular budget. With $\alpha = 0.5$ (a 26.6° tangent rotation), the actual next-token hidden state moves 57°–69° away from the original state, i.e., roughly $2\times$ – $2.6\times$ the controlled step. This means the natural dynamics do not stay on the shortest controlled arc; they jump to a different region of the sphere. The spherical geodesic is therefore a useful geometric idealization, not a literal description of the model’s own token-by-token trajectory.

3.5 Quantitative Traversal

Table 1 gives corrected rank-1 rates under the target tangent and fixed-score controls for five model families run with identical settings (64 targets, 2 contexts, 4 depth-adaptive layers, seed 42, fp16).

Model	Arc-reach	Target	Wrong	Random	Toward	Away
Qwen2-1.5B	99.2%	99.2%	1.6%	0.0%	34.8%	100.0%
Qwen2-0.5B	97.3%	97.3%	1.6%	0.0%	13.7%	100.0%
GPT-2	90.8%	90.8%	1.0%	0.0%	5.9%	99.2%
SmolLM-135M	67.2%	67.2%	0.4%	0.0%	2.5%	96.7%
Pythia-160M	27.9%	27.9%	0.4%	0.0%	9.0%	70.1%

Table 1: Rank-1 rates under corrected contracts. Wrong-target is self-excluded; random is a random tangent; toward/away are fixed-score residual rotations pointing at or away from the strongest competitor. Arc-reach = fraction of cases where the target-row arc reaches rank 1 within a 17° budget.

The rank improvement along the target tangent is guaranteed ($W_t \cdot g_t = \|g_t\|^2 \geq 0$). The fraction reaching rank 1 is high for the Qwen family and GPT-2, moderate for SmolLM-135M, and lower for Pythia-160M, showing that the phenomenon is robust yet model-dependent. Across all 2,560 cases, any target that became rank 1 along the arc remained rank 1 at the arc endpoint; this “no mid-arc loss” property was also confirmed at a 45° budget.

Table 2 reports the first rank-1 angle distribution over reachable cases.

Model	Median	Q25	Q75	P90	Reachable
Qwen2-1.5B	8.00°	6.4°	9.9°	11.3°	99.2%
Qwen2-0.5B	10.60°	9.0°	12.2°	14.1°	97.3%
GPT-2	11.60°	1.3°	13.8°	15.4°	90.8%
SmolLM-135M	9.84°	6.2°	13.4°	15.3°	67.2%
Pythia-160M	2.64°	2.0°	3.7°	9.3°	27.9%

Table 2: First rank-1 angle distribution over reachable cases. GPT-2’s low Q25 reflects a subset of very easy tokens.

3.6 Qualitative Demonstration

On Qwen2.5-7B, steering open-ended prompts produces visible output shifts. At $\alpha = 2$ on “Tell me something interesting:”, steering toward “75” changes the continuation from a generic fact (“The most common blood type is O positive...”) to a numeric statement (“1984 was the last year the United States had a budget surplus...”). At $\alpha = 3$ on “Once upon a time”, steering toward “apple” turns the story from “a young boy named Timmy...” into “apple trees were planted in the garden...” Common words and digits shift more readily than rare tokens; larger α trades fluency for strength.

Steering does not distinguish censored from uncensored paths. It is a white-box traversal primitive: with access to the model weights, anyone can move h toward any token direction on the sphere without retraining or fine-tuning.

4 Cow Tipping

Some tokens generate themselves when fed repeatedly. A token T is *self-consistent* if $s(T) = \text{softmax}(W \cdot h_T)[T]$ is high, where $h_T = \text{model}(e_T)$. A *pit* satisfies both high $s(T)$ and trajectory stability $\cos(h_{n+1}, h_n) \approx 1$: indefinite self-generation.

We scanned a subset of the Qwen2.5-7B vocabulary, testing permanence (3 input repetitions, 15-step self-prediction). Representative tokens pass:

Token	ID	$s(T)$	Trigger
0	15	0.975	000, 000 – 000 – 0000
ab	370	0.934	ababab
. (space+dot)	659	0.889	. . .
ere	485	0.878	erere
????	26610	0.386 [†]	8+ question marks

[†]Deterministic fixed point ($\cos \approx 1.0$ once generated).

The 0 token is clearest: 000 yields $p = 0.974$, $\cos \approx 1.0$, $p > 0.95$ within 3 steps. Phone numbers trigger it via three consecutive token-15 units. Because pits are derived from model weights, each model’s pits are unique to its tokenizer and LM head geometry.

4.1 Decoder Contract

The original measurement helper used two non-standard decoder choices: temperature only rescaled logits before ‘argmax’ (so $T = 0.8$ was still greedy), and top-p sampling was uniform over the retained nucleus rather than the standard probability-weighted renormalized nucleus. Under the corrected decoder contracts, the 0.5B “0” loop is strongly decoder-dependent, while the Qwen2.5-7B strict “0” pit is much more robust.

Decoder	Qwen2-0.5B mean trailing-0	Qwen2.5-7B mean trailing-0
Greedy	24.0	30.0
Multinomial $T = 0.8$	1.12	28.83
Top-p 0.9, $T = 1.0$ weighted	0.78	30.0
Top-p 0.9, $T = 0.8$ weighted	2.47	30.0
Top-p 0.9, $T = 0.8$ uniform	0.16	30.0
Chat-template wrapped	0	0

On Qwen2.5-7B the strict pit survives greedy decoding and both standard and uniform nucleus sampling; only the chat-template context breaks it. On Qwen2-0.5B the loop is a degenerate repetition basin that sampling usually disrupts. The difference is important: the 0.5B behavior cannot be assumed to generalize to 7B strict pits, and the decoder contract must be reported with every measurement.

4.2 Threat Model and Mitigations

Attacker capabilities. An attacker with white-box access (or API access and a vocabulary scan) can find high- $s(T)$ pit tokens for a target model, then embed those tokens at chunk boundaries in data the model will process. The result is a fixed-point loop that degrades output quality or causes denial of service.

Attack scenarios.

- **Scraping degradation:** a web page terminates in a pit trigger; an LLM-based crawler loops instead of summarizing.
- **Prompt injection:** a user includes a short trigger string to make the assistant loop.
- **Compute exhaustion:** forcing long repeated-token generation increases inference cost.

Mitigations. We evaluated eight classes of defense on Qwen2-0.5B and Gemma-3-1B, measuring loop-break rate on the discovered pit trigger and false-positive rate (median truncation) on 20 ordinary prompts. Table 3 summarizes the most reliable ones.

On Gemma-3-1B the discovered pit token is <mask> (id 4) with a greedy loop length of 35. The same practical defenses break it: repetition detector, output collapse, multinomial/top-p sampling, and pit-away steering all reduce the loop to 0–10 tokens with no false positives. Entropy monitoring does not fire on the

Mitigation	Qwen2-0.5B loop break	Median FP truncation
Repetition detector (≥ 3 same pit)	yes (loop 3)	0 tokens
Repetition detector (≥ 4 same pit)	yes (loop 4)	0 tokens
Output collapse repeats	yes (loop 2)	0 tokens*
Entropy floor 1.0 + pit penalty	yes (loop 1)	0 tokens
Pit-away steering (last layer, $\alpha = 0.3$)	yes (loop 0)	0 tokens
N-gram detector (4-gram, ≥ 2)	yes (loop 8)	23 tokens
Periodicity detector (≥ 0.85)	no	24 tokens

Table 3: Qwen2-0.5B mitigation results. *Output-collapse false positives were not measured separately because the mitigation is applied only after generation halts.

Gemma pit (the output distribution is not extremely concentrated), illustrating that no single detector is universal.

Repetition detection, output collapse, entropy monitoring, and pit-away steering break the loop with no measurable false positives on ordinary generation in this small set. The detector can be deployed either as an input filter (collapsing repeated pit tokens) or as an output guard (halting generation once N identical tokens are emitted). N-gram and periodicity detectors also break the loop but generate unacceptable false positives on normal text. Input sanitization alone is less reliable: stripping repeated pit tokens reduces the loop length but does not always break it. Decoder changes (higher temperature, standard top-p) are effective on shallow basins but may not break strict 7B pits; for strict pits, input/context defenses and pit-aware steering are more reliable.

Responsible framing. Pits are a model-weight-derived phenomenon, so any mitigation must be calibrated per deployment model. The defensive-encoding use case is one application, but the same mechanism can be used for prompt injection or scraping degradation. We report it as a capability that requires careful deployment, not as a universal defense.

4.3 Defensive Encoding

The accompanying code release ([steering-evals/scripts/](#)) provides a full scan that finds pits for any model. Embedding pit triggers into data causes a model that processes it to fall into a fixed-point loop rather than continue reading. Because the basins are narrow, the trigger must be the terminal content of each chunk. Verified encodings include NULL bytes, repeated sub-tokens, and terminal phone-number-like strings. A crawler reading a page that terminates in one of these triggers stops producing useful output and repeats the pit token instead.

5 Edge of Chaos: A Theory

The residual stream is a dynamical system $h_{l+1} = h_l + f_l(h_l)$. Its per-layer Lyapunov exponent measures how perturbations grow:

$$\lambda_l = \ln \frac{\|h_{l+1}^{\text{pert}} - h_{l+1}^{\text{ref}}\|}{\|h_l^{\text{pert}} - h_l^{\text{ref}}\|}$$

A perturbation that neither explodes nor vanishes during computation would place the system at the edge of chaos, corresponding to $\lambda \cdot L_{\text{comp}} \approx 0.5$ (one e-fold of total expansion). We measured the per-layer rate across 13 model families:

Cluster	Models	$\lambda \cdot L_{\text{comp}}$
≈ 0.5	Gemma-2-2B, Gemma-2-9B, GPT-2, DS-R1-7B, Qwen-1.5B, Qwen2.5-7B	0.37–0.57
≈ 0.8	SmolLM2-360M, DeepSeek-7B	0.76–0.88
≈ 1.0	Phi-3.5-mini, Mistral-7B	1.00–1.47
≈ 0.1	Qwen2.5-0.5B, Qwen2.5-3B	0.01–0.10

The edge-of-chaos balance ($\lambda \cdot L \approx 0.5$) holds for roughly half the models—notably the Gemma family, GPT-2, and mid-sized Qwen—but is not universal. We present it as a theory that captures a subset of architectures, not as a scaling law. Cow tipping (Section 4) is the degenerate limit of this balance: a fixed point where $\lambda_l \rightarrow 0$ and the trajectory stops changing entirely.

6 Reproduction

All measurements are scriptable in under 100 lines of Python. Full code at <https://github.com/ntrillard/transformer-geometry>. The corrected measurement harness lives in `steering-evals/scripts/` and implements the self-excluding wrong-target control, residual-latitude off-arc rotations, raw-head rank-cone projection with NNLS dual, and the corrected decoder samplers. The procedure:

1. **Sphere radius.** Extract γ_l from model state dict. Feed prompts, record post-LN hidden state norms per layer. Compare to $\|\gamma_l\|$.
2. **BOS axis.** Feed a prompt, extract hidden states per layer. For each layer, compute $\hat{b}_l$ from position 0 and $\theta_{l,t}$ for any LM head row W_t .
3. **Steering (traversal).** Forward-pass a prompt to get h . For target token t , compute g_t , step with α , renormalize by $\|\gamma_L\|$, then feed h' as the first-generation hidden state. Use `steering_geometry_test.py` for corrected specificity, competitor geometry, and shortest-route measurements.
4. **Cow tipping.** For token t , decode, re-encode, forward-pass to get h_T . Compute $s(T)$. If $s(T) > 0.01$, feed t three times and check 15-step self-prediction with greedy decoding. Record permanence and cos stability. Use `eval_pit_robustness.py` and `eval_7b_strict_pit_gate.py` for decoder-contract tests, and `eval_threat_model.py` for mitigation evaluation.

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References

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